Case Study 1: Cyclistic Bike-Share Analysis

Executive Summary

This case study analyzes 12 months of Cyclistic bike-share data (February 2025–January 2026) to identify behavioral differences between annual members and casual riders. The objective is to support data-driven marketing strategies aimed at increasing annual memberships.

Using Google BigQuery for data cleaning and transformation and Tableau Public for visualization, key usage patterns were identified across ride frequency, duration, weekday behavior, and seasonal trends.

Results show that annual members account for 64% of total rides (3.55M rides) and demonstrate consistent weekday usage patterns indicative of commuting behavior. Casual riders, by contrast, take longer trips on average (22.63 minutes vs. 12.41 minutes), show stronger weekend activity, and display greater seasonal variation.

These findings highlight targeted opportunities to convert casual riders into annual members through cost-efficiency messaging, weekend-focused promotions, and seasonal pre-summer campaigns.

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1. Ask

Business Task

The objective of this analysis is to identify behavioral differences between annual members and casual riders in order to support data-driven marketing strategies aimed at converting casual riders into annual members.

Cyclistic's leadership believes that increasing annual memberships is key to long-term growth and profitability. Therefore, understanding usage patterns across rider segments is essential to inform targeted marketing initiatives.

Key Stakeholders

- **Lily Moreno** – Director of Marketing
- **Cyclistic Executive Team** – Responsible for approving marketing strategies
- **Cyclistic Marketing Analytics Team** – Responsible for data analysis and reporting

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Guiding Questions

- How do annual members and casual riders use Cyclistic bikes differently?
- Why might casual riders consider purchasing an annual membership?
- How can Cyclistic use digital marketing strategies to influence casual riders to become annual members?

Deliverable

A clear, data-driven analysis identifying usage patterns between casual riders and annual members, supported by visual dashboards and actionable marketing recommendations.

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2. Prepare

Data Source

The dataset consists of Cyclistic's historical trip data covering a 12-month period (February 2025–January 2026). The data was made publicly available by Motivate International Inc. and does not contain personally identifiable information (PII).

The original data was downloaded as 12 separate monthly CSV files. These files were uploaded to Google Cloud Storage and imported into Google BigQuery for processing and analysis.

Data Organization

Each monthly file contained ride-level records including:

- ride_id
- rideable_type
- started_at
- ended_at
- start_station_name
- end_station_name
- start_lat / start_lng
- end_lat / end_lng
- member_casual

After verifying structural consistency across all files, the 12 datasets were consolidated into a single BigQuery table for full-year analysis.

Data Credibility (ROCCC)

The dataset meets the ROCCC criteria:

- **Reliable** – Provided by a reputable bike-share operator
- **Original** – First-party operational trip data
- **Comprehensive** – Includes a full 12-month period
- **Current** – Reflects recent rider behavior
- **Cited** – Publicly available under appropriate licensing

Privacy and Security

The dataset contains no sensitive customer information such as names, addresses, or payment details. The analysis focuses strictly on ride behavior patterns.

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3. Process

Tools Used

- **Google BigQuery (SQL)** – Data cleaning and transformation
- **Tableau Public** – Data visualization and dashboard creation

BigQuery was selected for its ability to efficiently process large datasets and perform structured queries.

Data Cleaning Steps

After consolidating all 12 months of data into one table, the following cleaning steps were performed:

- Removed rows with missing timestamps (started_at or ended_at)
- Removed invalid ride durations where end time was before or equal to start time
- Created a new column: **ride_length** (calculated from ended_at – started_at)
- Created a new column: **day_of_week** (derived from started_at)
- Created a cleaned working table to preserve the original raw dataset

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Data Validation

Post-cleaning validation included:

- Comparing row counts before and after cleaning
- Verifying column consistency
- Confirming complete monthly coverage
- Reviewing ride duration calculations

The final cleaned dataset was then exported to Tableau Public to develop interactive dashboards for stakeholder review and executive decision-making.

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4. Analyze

Summary of Analysis

Descriptive analysis was conducted to identify behavioral differences between annual members and casual riders, focusing on total ride frequency, average ride length, weekday usage patterns, and monthly trends.

Total Ride Frequency

Annual members recorded **3,550,958 rides**, while casual riders recorded **1,999,301 rides** during the 12-month period.

Annual members account for **64% of total rides**, significantly outpacing casual riders in overall frequency.

This indicates that members use Cyclistic bikes more consistently and frequently.

Average Ride Length

- Annual Members: **12.41 minutes**
- Casual Riders: **22.63 minutes**

Casual riders take substantially longer trips on average. This suggests members are likely using bikes for short, routine trips such as commuting, while casual riders are using bikes for leisure or recreational purposes.

Weekly Usage Patterns

Distinct behavioral patterns emerge across the week:

- Members show consistently high ride volume during weekdays (Monday–Friday)
- Member ride counts decrease during weekends
- Casual riders show elevated activity during weekends, especially Saturdays

Although members maintain higher overall volume, casual rider activity increases significantly on weekends compared to their weekday levels.

This pattern suggests commuting behavior among members and leisure-driven usage among casual riders.

Monthly Trends

Monthly analysis reveals additional segmentation:

- Member ride volume remains relatively stable throughout the year
- Casual rider activity shows strong seasonal variation
- Casual ridership increases during warmer months and declines in colder months

The stability of member ridership suggests habitual commuting behavior, while the seasonal variability of casual riders indicates discretionary, leisure-oriented usage influenced by weather conditions.

Key Insight

Annual members demonstrate frequent, short-duration, weekday-oriented riding behavior consistent with commuting.

Casual riders demonstrate longer ride durations, stronger weekend engagement, and seasonal sensitivity consistent with leisure and recreational usage.

These behavioral differences provide clear segmentation between rider types.

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5. Share

Data Visualizations

To communicate findings effectively, interactive dashboards were developed in Tableau Public including:

- **Total Rides by Rider Type (Column Chart)**
- **Average Ride Length by Rider Type (Column Chart)**
- **Rides by Day of Week and Rider Type (Clustered Column Chart)**
- **Monthly Ride Trends by Rider Type (Line Chart)**

These visualizations were consolidated into an interactive dashboard, enabling stakeholders to dynamically explore rider behavior patterns across frequency, duration, seasonality, and weekday usage.

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6. Act

Strategic Recommendations

1. Target High-Activity Weekend Casual Riders

Since casual riders demonstrate elevated weekend activity, targeted digital campaigns during peak weekend periods should promote annual memberships with limited-time discounts or upgrade incentives.

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2. Emphasize Cost Efficiency for Frequent Long-Duration Riders

Given that casual riders average longer ride durations, marketing materials should highlight cost savings associated with unlimited or discounted rides under annual membership plans.

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3. Launch Pre-Summer Conversion Campaigns

Because casual ridership increases during warmer months, promotional campaigns should be launched in early spring to convert high-intent seasonal riders before peak summer demand.

Conclusion

This analysis demonstrates clear behavioral differences between annual members and casual riders. Members exhibit consistent weekday usage and shorter ride durations aligned with commuting patterns. Casual riders display longer ride durations, increased weekend activity, and greater seasonal sensitivity indicative of leisure-based usage.

By leveraging these insights, Cyclistic can implement targeted marketing strategies to convert casual riders into more profitable annual members, supporting long-term growth and revenue stability.

Tools & Technologies Used

• Google BigQuery (SQL) – Data cleaning and transformation
• Google Cloud Storage – Data storage and file management
• Tableau Public – Data visualization and dashboard creation
• Microsoft Word
• Generative AI tools – Assisted with query refinement and report drafting (outputs reviewed and validated)